Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: F&S Brasil #2

Physical address of dairy:

Number and Street	City	County	Zip Code
6052 Trahern RD	Manteca	San Joaquin	95337

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1930

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X257-X110-XX16-XXXX

B. OPERATORS

Brasil, Frank A

Operator name: Brasil, Frank A

Telephone no.: Landline: ; Cellular: (209) 678-0903

Mailing Address Number and Street	City	State	Zip Code
6524 E Perrin RD	Manteca	CA	95337

C. OWNERS

Hofman, Dirk

Legal owner name: Hofman, Dirk

Telephone no.: Landline: (209) 239-2321; Cellular:

Mailing Address Number and Street	City	State	Zip Code
7001 E Trahern RD	Manteca	CA	95337

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	72	69	72	54	24	18
Number under roof	0	0	0	0	0	0
Maximum number	240	229	250	200	100	90
Average number	72	69	72	54	24	18
Avg live weight (lbs)	1,250	1,300	950	750		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 65 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 4,177.58 *tons per reporting period*

Total nitrogen from manure: 50,291.06 *lbs per reporting period*

After ammonia losses (30% loss applied): 35,203.74 *lbs per reporting period*

Total phosphorus from manure: 7,823.97 *lbs per reporting period*

Total potassium from manure: 13,552.14 *lbs per reporting period*

Total salt from manure: 49,767.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Stanislaus Water Users Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	34	34	0	none	X257-XX11-X001-XXXX
Field 2	53	53	0	none	X257-XX11-X001-XXXX
Field 3	44	44	1	none	X257-XX11-X001-XXXX
Field 4	32	32	1	none	X257-XX11-X016-XXXX
Field 5	32	32	1	none	X257-XX11-X017-XXXX
Totals for areas that were used for application					
Totals for areas that were not used for application	195	195	3		
Land application area totals	195	195	3		

B. CROPS AND HARVESTS

Field 3

Field name: Field 3

01/27/2019: Almonds Field Run

Crop: Almonds Field Run Acres planted: 44 Plant date: 01/27/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/15/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Field 4

Field name: Field 4

Field 4

01/01/2023: Pasture

Crop: Pasture Acres planted: 32 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/15/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

Field 5

Field name: Field 5

11/14/2013: Almonds Field Run

Crop: Almonds Field Run Acres planted: 32 Plant date: 11/14/2013

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/15/2023	0.10 *ton*	As-is		0.1	0.00	0.00	0.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	0.00	0.00	0.00	0.00	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 3 - 01/27/2019: Almonds Field Run

Field name: Field 3

Crop: Almonds Field Run — Plant date: 01/27/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	32.08	4,698,956.00 *gal*
Application event totals		0.00	0.00	0.00	32.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	29.08	4,259,646.00 *gal*
Application event totals		0.00	0.00	0.00	29.08	

Field 4 - 01/01/2023: Pasture

Field name: Field 4

Crop: Pasture — Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	51.26	5,460,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.26	

Field 5 - 11/14/2013: Almonds Field Run

Field name: Field 5

Crop: Almonds Field Run — Plant date: 11/14/2013

Field 5 - 11/14/2013: Almonds Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	30.59	3,258,896.00 *gal*
Application event totals		0.00	0.00	0.00	30.59	

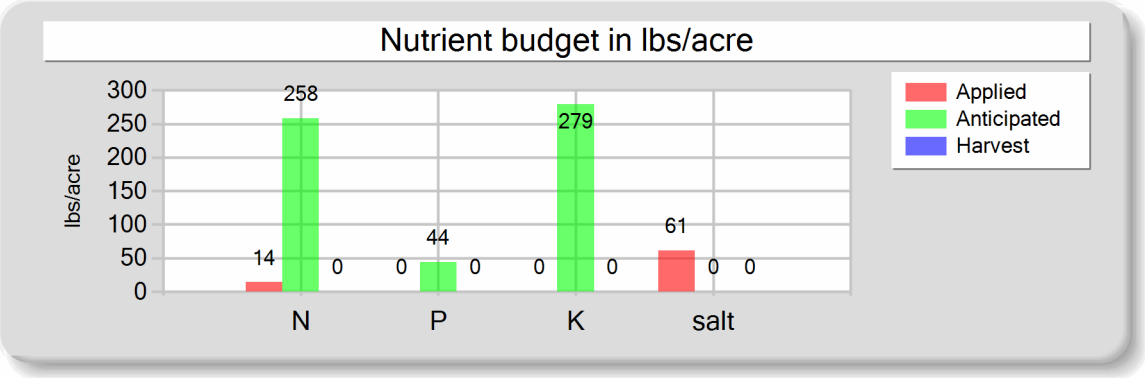
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Stanislaus Water Users Canal	Surface water	0.00	0.00	0.00	28.75	3,062,663.00 *gal*
Application event totals		0.00	0.00	0.00	28.75	

B. NUTRIENT BUDGET

Field 3 - 01/27/2019: Almonds Field Run

Field name: Field 3 Crop: Almonds Field Run Plant date: 01/27/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	61.17
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	61.17
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	14.00	0.00	0.00	61.17
Applied to removed ratio	0.00	0.00	0.00	0.00

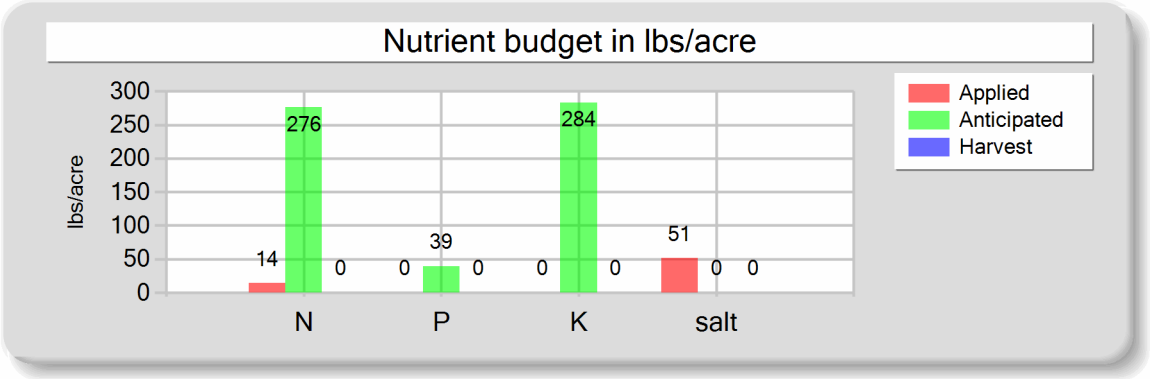
Fresh water applied	
8,958,602.00	*gallons*
329.91	*acre-inches*
7.50	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 01/01/2023: Pasture

Field name: Field 4 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	51.26
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	51.26
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	14.00	0.00	0.00	51.26
Applied to removed ratio	0.00	0.00	0.00	0.00

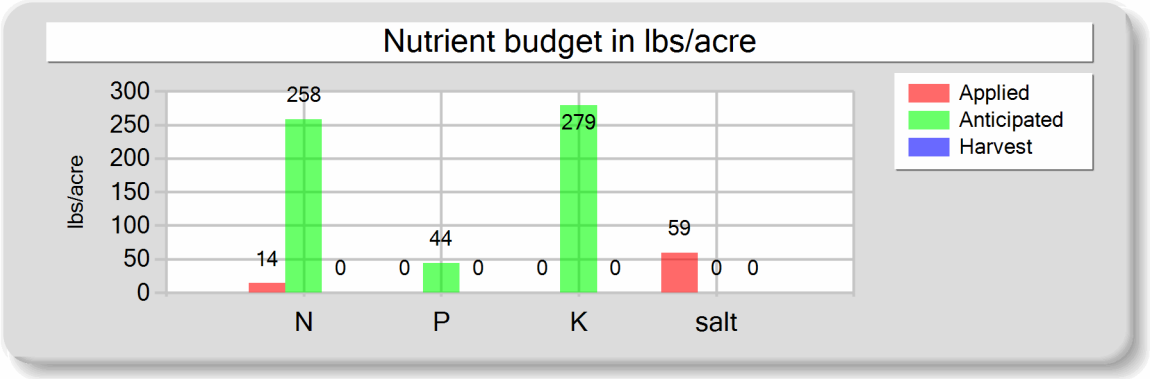
Fresh water applied	
5,460,000.00	*gallons*
201.07	*acre-inches*
6.28	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/14/2013: Almonds Field Run

Field name: Field 5 Crop: Almonds Field Run Plant date: 11/14/2013



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	59.35
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	59.35
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	14.00	0.00	0.00	59.35
Applied to removed ratio	0.00	0.00	0.00	0.00

Fresh water applied	
6,321,559.00	*gallons*
232.80	*acre-inches*
7.28	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

No manure analyses entered.

B. PROCESS WASTEWATER ANALYSES

1st Qtr WW

Sample and source description: 1st Qtr WW

Sample date: 02/03/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	367.00	67.20	0.00	0.00	62.20	460.00								2,690.00	2,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Stanislaus Water Users Canal

Stanislaus Water Users Canal

Sample description: Stanislaus Water Users Canal

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								65.00	36
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

No plant tissue analyses entered.

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

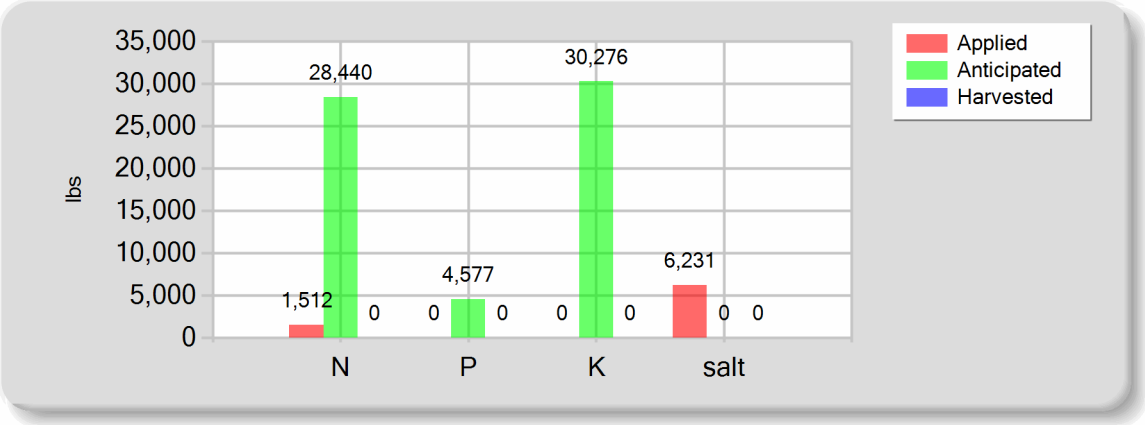
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

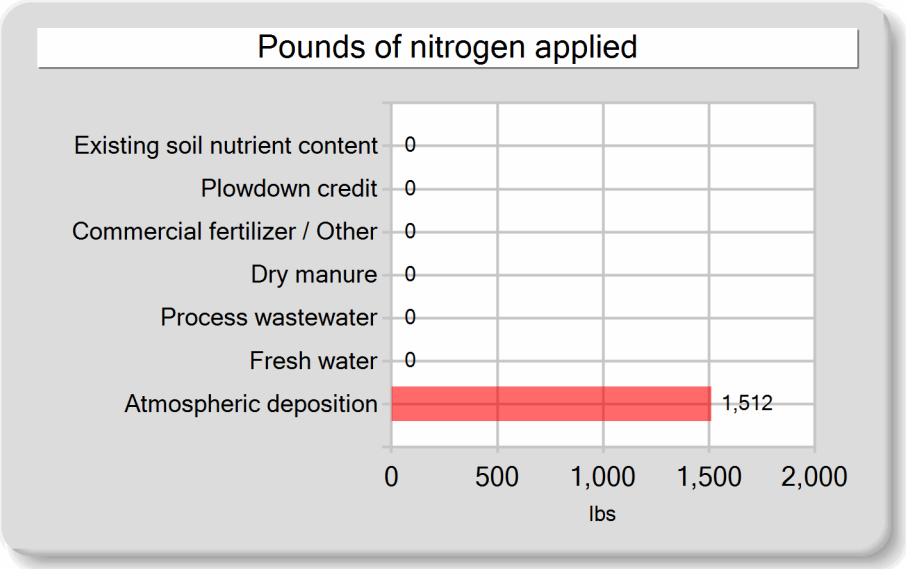
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

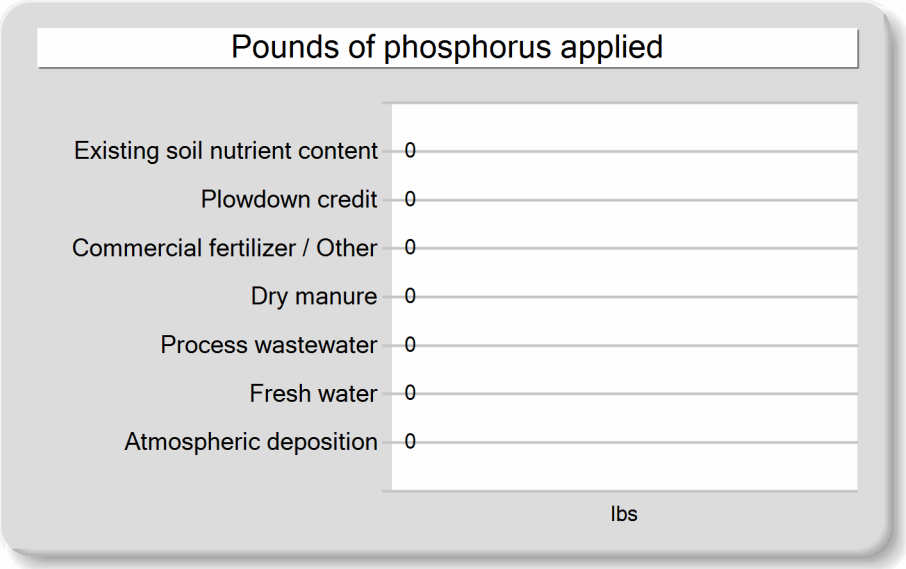
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	6,230.76
Atmospheric deposition	1,512.00	0.00	0.00	0.00
Total nutrients applied	1,512.00	0.00	0.00	6,230.76
Anticipated crop nutrient removal	28,440.00	4,576.80	30,276.00	0.00
Actual crop nutrient removal	0.00	0.00	0.00	0.00
Nutrient balance	1,512.00	0.00	0.00	6,230.76
Applied to removed ratio	0.00	0.00	0.00	0.00

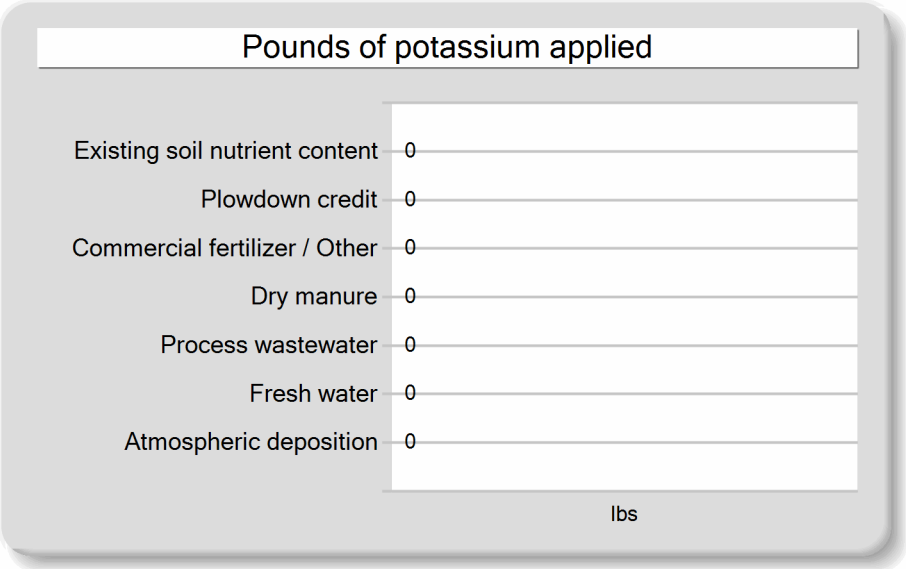
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

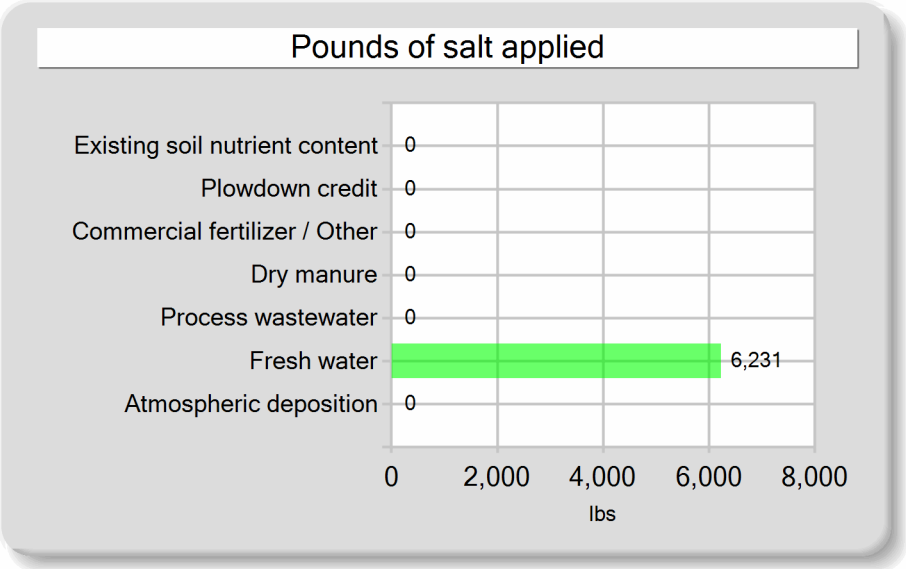


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

This Annual Report submission represents the period of 1/1/2023 through 4/15/2023 when the operator ceased operations at this facility. Land applications and Harvest Yields that occurred after 4/15/2023 were unavailable for the purpose of this report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Dirk Hofman	Frank A Brasil
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/24/24	6-24-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Brasil/Machado (F & S Brasil)
6524 Perrin Rd
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #2
RECEIVED DATE: 3/20/23 7:30 AM
SUBMITTED BY: Sean
ANALYZED DATE: 3/20/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Milk Barn	T3079006A	Electrical Conductivity (EC)	EPA 120.1 and	0.582	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.35	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	372	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	104	93.2 (T3079001A)	93.3 (T3079001A)	0.064
Electrical Conductivity (EC)	96.4			0.000
Nitrate Nitrogen (NO3-N)	104	102 (T3076025A)	101 (T3076025A)	1.52

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Brasil/Machado (F & S Brasil)
6524 Perrin Rd
Manteca, CA 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #2
RECEIVED DATE: 3/20/23 7:30 AM
SUBMITTED BY: Sean
ANALYZED DATE: 3/20/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Houses	T3079006B	Electrical Conductivity (EC)	EPA 120.1 and	0.439	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.485	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	281	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	104	93.2 (T3079001A)	93.3 (T3079001A)	0.064
Electrical Conductivity (EC)	96.4			0.000
Nitrate Nitrogen (NO3-N)	104	102 (T3076025A)	101 (T3076025A)	1.52

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: F+S BRASIL
Mailing address:
(not facility address) MANTECA CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID: DAIRY #2

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	3/17/23	10:45	DW- MILK BARN	DOMESTIC	3.6°C		X				
2	"	10:50	DW- HOUSES	"	3.0°C		X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	3-20-23 7:30			T3079006

Required to be submitted to any agency? (YES / NO)